

1. Introduction

Phishing is a type of social engineering attack where attackers attempt to trick individuals into revealing sensitive information such as usernames, passwords, credit card details, or other personal data. These attacks are usually carried out through fraudulent emails, messages, or fake websites that appear to be from legitimate and trusted sources such as banks, social media platforms, or online services. The primary goal of phishing attacks is to exploit human psychology rather than technical vulnerabilities. Attackers often create a sense of urgency, fear, or curiosity to manipulate users into taking immediate action, such as clicking a malicious link or entering their credentials on a fake website. As a result, even well-designed security systems can fail if users are not aware of such threats.

Phishing attacks can lead to serious consequences, including identity theft, financial loss, unauthorized access to accounts, and data breaches. With the increasing dependency on online services, it has become essential to educate users about these threats and how to identify and avoid them. This project focuses on simulating a phishing attack in a controlled and ethical environment. A fake email and a login page are created to mimic a real-world phishing scenario. The purpose of this simulation is to observe user behavior, measure awareness levels, and understand how easily individuals can be deceived.

2. Methodology

Step 1: Research

Different phishing techniques such as email phishing and fake login pages were studied.

Step 2: Email Design

A fake phishing email was created with an urgent message asking users to reset their password.

Step 3: Fake Login Page

A simple HTML-based login page was developed to mimic a real website login interface.

Step 4: Simulation

The email was shared with a small group of participants (classmates/friends) after informing them that it was part of a simulation.

Step 5: Data Collection

User responses were recorded based on:

- Whether they clicked the link
- Whether they entered data
- Whether they identified it as phishing

3. Results

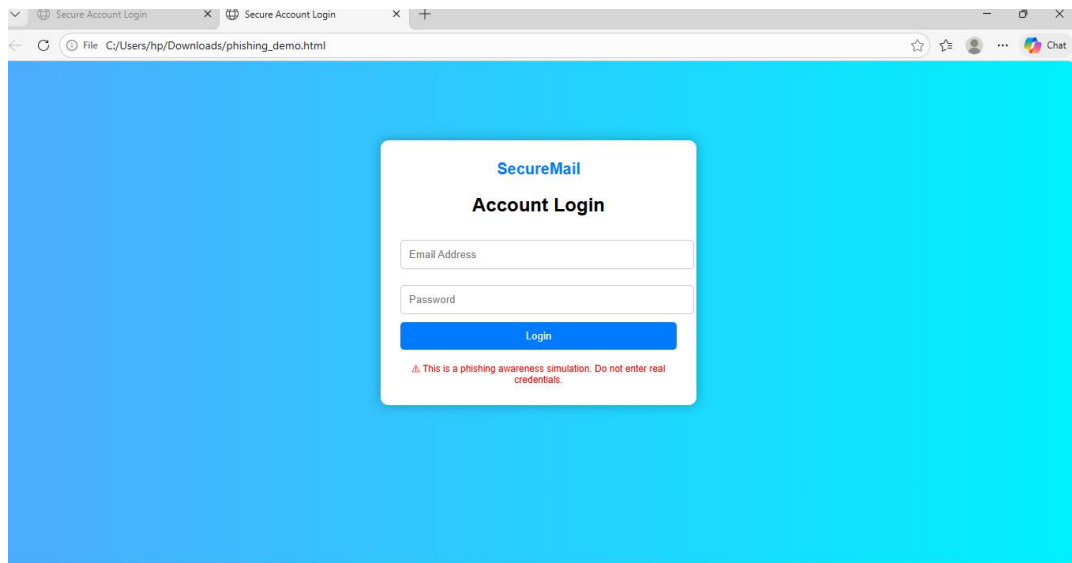


Fig 1: HTML fake login page

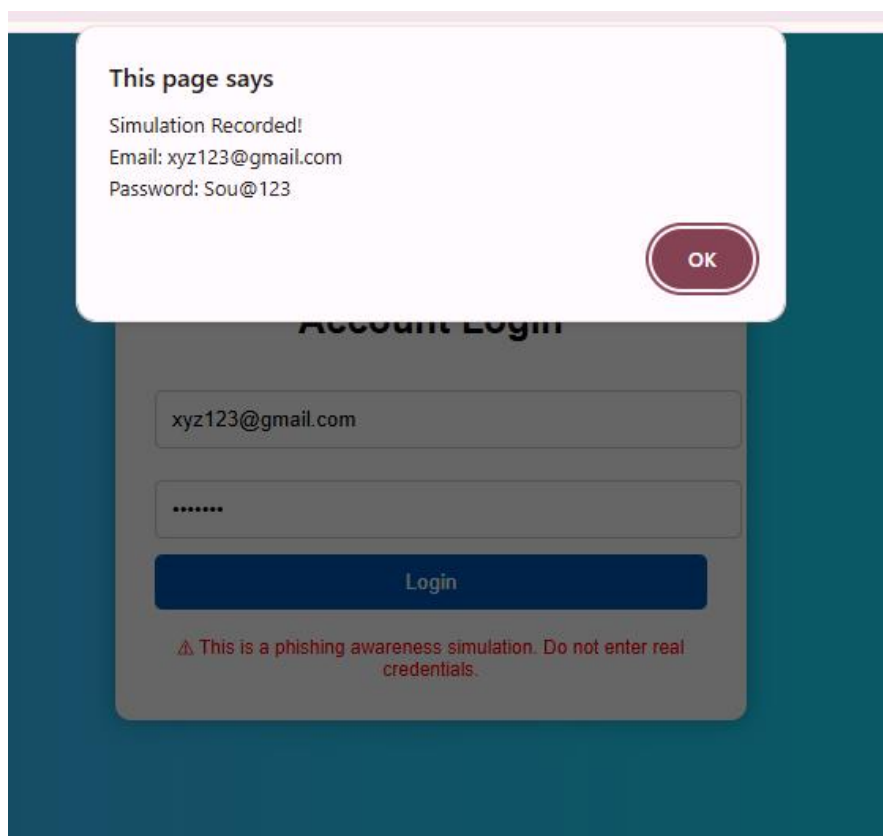


Fig 2: Working simulation (popup capturing input)



Fig 3: Phishing email simulation on screen

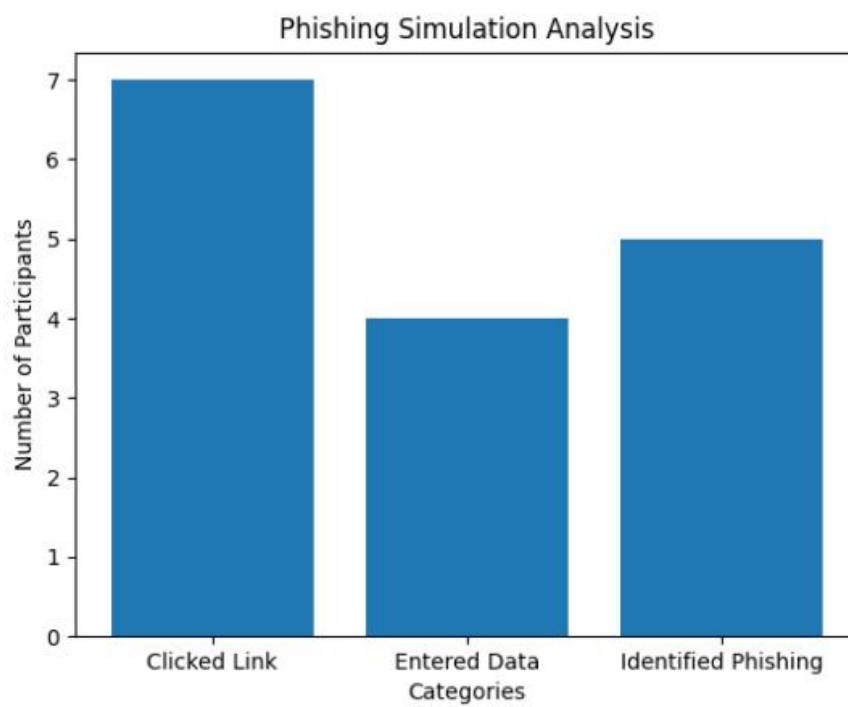


Fig 4: Click states of Phishing Simulation Analysis

4. Lessons Learned

- Users tend to trust emails that appear urgent or official
- Lack of cybersecurity awareness increases vulnerability
- Even simple phishing techniques can be effective
- Awareness and training significantly reduce risk
- Users should always verify the authenticity of emails before taking action

5. Preventive Measures

➤ Technical Measures

- Use spam filters to detect phishing emails
- Enable Multi-Factor Authentication (MFA)
- Use secure browsers and antivirus software

➤ User Awareness

- Avoid clicking on unknown or suspicious links
- Verify the sender's email address
- Check for spelling mistakes and unusual requests
- Never share sensitive information through email

➤ Organizational Measures

- Conduct regular phishing awareness training
- Implement strong cybersecurity policies
- Encourage reporting of suspicious emails